

Jaya Agarwal

✉ jayagarwalfd@gmail.com ☎ 8826840373  [linkedin.com/in/jaya723](https://www.linkedin.com/in/jaya723)  github.com/Jaya280

Summary

Enthusiastic Computer Science student specializing in Artificial Intelligence and Machine Learning with hands-on experience in LLM applications, RAG systems, deep learning, machine learning, data analysis, databases, and Python Flask development. Passionate about building practical AI-powered solutions and solving real-world problems using modern GenAI technologies.

Education

Manipal University Jaipur (CGPA: 9.8/10 (till 5th semester))

1. B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning)
2. *Dean's Excellence Award recipient for five consecutive semesters (CGPA above 9.0)*

Class 12(CBSE Board): 87%

- Head Girl / Student Council Leader (11th grade)

Class 10(CBSE Board): 94.4%

Skills and Technologies

Programming: Python (OOPS), C++ (DSA)

Core CS: Operating Systems, RDBMS, Computer Networks, OOPS

Software Development: Flask (REST APIs, JWT authentication), Debugging & Testing (Swagger API Docs, SonarQube)

Data Science, AI & ML: Pandas, NumPy, Matplotlib, Seaborn, Scikit Learn, Data Cleaning, Supervised Learning Algorithms, Deep learning, Tensorflow, Keras, Generative AI, RAG, Langchain

Tools & Technologies: Git, GitLab, GitHub, Visual Studio Code, Jupyter Notebook, Google Colab, QGIS

Experience

Software Development Intern(Techfour Engineering Solutions, May - July 2025)

- Developed a modular microservices-based system (Document, Communication, User Management) to streamline internal workflows and automate routine business processes.
- Built the backend using Python Flask and frontend with Flutter, integrating APIs and implementing JWT-based authentication for secure access.
- Documented APIs with Swagger, maintained code quality using SonarQube, and followed Agile methodology with Git/GitLab for version control.

Projects

LULC Change Detection using Deep Learning

AI Powered Solution Expo Winner

- Built an end-to-end LULC change detection pipeline using PlanetScope 3m imagery, QGIS preprocessing, Random Forest classification, and deep learning models.
- Compared Siamese U-Net and BiUNet-Dense architectures for bi-temporal urban change detection, achieving 92.25% accuracy and 0.8673 F1-score.
- Generated urban expansion and land-cover change maps for Chandigarh (2016–2025) for geospatial analysis and planning insights.

Gurgaon Real-Estate

[Click here](#)

- Developed an XGBoost-based house price prediction system using engineered features from unstructured real-estate data.
- Designed a weighted hybrid recommender system using Cosine Similarity for personalized property recommendations.
- Built a multi-page Streamlit dashboard for real-time prediction, analytics, and recommendations.

Certificates

- **Python Essentials 1 & 2**, Cisco— Core Python programming.
- **Programming, Data Structures and Algorithms using Python**, NPTEL- Algorithms data structures in Python.
- **Design and Analysis of Algorithms**, NPTEL— Algorithm efficiency optimization.
- **Introduction to Machine Learning**, NPTEL - ML fundamentals and model evaluation.